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Decoding cardiovascular risk in Chinese middle-aged and elderly adults: a 9-year prospective study integrating machine learning with explainable AI based on CHARLS cohort

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Abstract

Background Cardiovascular disease constitutes the most formidable public health challenge in China, accounting for 48.98% and 47.35% of mortality in rural and urban populations, respectively, affecting approximately 330 million individuals. Existing risk stratification models predominantly derive from Western populations, with the Framingham Risk Equation systematically overestimating cardiovascular risk by 276% in Chinese men and 102% in Chinese women, underscoring the critical imperative for population-specific predictive instruments. Although machine learning methodologies demonstrate considerable promise in cardiovascular risk prognostication, their inherent "black-box" characteristics substantially impede clinical translational implementation.

Objective Leveraging longitudinal cohort data from the China Health and Retirement Longitudinal Study (CHARLS) and integrating machine learning with explainable artificial intelligence techniques, we sought to develop and validate a cardiovascular disease long-term risk prediction model tailored to the Chinese middle-aged and elderly population, achieving optimal synthesis of predictive accuracy and clinical interpretability through quantitative risk factor contribution analysis.

Methods We incorporated four waves of CHARLS surveillance data spanning 2011–2020, with 8,080 participants aged ≥ 45 years completing 9-year follow-up after rigorous inclusion criteria application. Recursive feature elimination was employed to identify optimal predictors from 90 candidate variables. We systematically evaluated 12 machine learning algorithms encompassing linear, non-linear, ensemble learning, and deep learning methodologies, utilizing stratified random 7:3 partitioning for training and validation cohorts. SHAP (SHapley Additive exPlanations) methodology facilitated comprehensive global and local interpretability analyses, with decision curve analysis assessing clinical net benefit.

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Results Among 5,699 training cohort participants, 1,248 (21.9%) experienced cardiovascular events during follow-up. Recursive feature elimination identified 18 pivotal predictive factors spanning lipid metabolism, anthropometric parameters, renal function, and glucose homeostasis domains. The gradient boosting machine demonstrated superior comprehensive performance, achieving validation cohort AUC of 0.798 (95% CI: 0.776–0.820), specificity of 98%, and positive predictive value of 78%. SHAP analysis revealed waist circumference, triglycerides, and hypertension history as the three predominant predictive factors, with mean absolute SHAP values significantly exceeding other variables. Individual risk attribution analysis demonstrated substantial heterogeneity: extremely high-risk specimens (predicted probability 0.991) exhibited synergistic multi-factorial risk amplification, with standardized waist circumference contributing + 0.0778 SHAP value and triglycerides (477 mg/dL) contributing + 0.0729; conversely, low-risk specimens (predicted probability − 0.0393) demonstrated triglycerides (45.1 mg/dL) providing the maximal singular protective contribution of -0.166. Decision curve analysis confirmed positive net benefit across the 0-0.95 threshold probability spectrum, systematically surpassing conventional strategies.

Conclusions The gradient boosting machine model achieved superior discrimination (AUC 0.798, 95% CI 0.785–0.825) compared to Framingham (0.638) and China-PAR (0.654) scores for 9-year cardiovascular disease prediction in Chinese adults aged ≥ 45 years. Waist circumference, triglycerides, and hypertension emerged as principal predictive features, though SHAP-derived importance reflects statistical contribution rather than causal effects. Decision curve analysis demonstrated clinical utility across threshold probabilities 0.05–0.95, enabling flexible deployment from population screening (98.3% sensitivity) to targeted intervention (98.7% specificity). External validation in independent cohorts is essential to establish generalizability before clinical implementation.

Clinical trial number Not applicable.

Keywords Cardiovascular disease prediction, Machine learning, Explainable artificial intelligence, SHAP analysis, Gradient boosting machine, Chinese middle-aged and elderly population, CHARLS cohort, Risk stratification

Introduction

Cardiovascular disease represents the preeminent global public health challenge, with particularly pronounced significance in China. Currently, cardiovascular diseases account for 48.98% and 47.35% of all-cause mortality in rural and urban Chinese regions, respectively, with approximately 330 million individuals nationwide affected by various cardiovascular pathologies [1]. Epidemiological projections indicate that cardiovascular disease incidence in China will escalate from 0.74% in 2021 to 0.97% by 2030 [2], a trajectory that underscores the critical imperative for refined risk stratification methodologies. However, contemporary cardiovascular risk prediction frameworks predominantly derive from Western cohorts, such as the Framingham Heart Study, exhibiting substantial limitations when extrapolated to Chinese populations. Evidence demonstrates that the Framingham risk equation systematically overestimates cardiovascular risk in Chinese individuals, with magnitudes of overestimation reaching 276% for males and 102% for females [3]. Furthermore, existing predictive algorithms developed from non-Asian populations demonstrate marked heterogeneity in performance across diverse Asian subpopulations [4]. This absence of validated long-term cardiovascular risk prediction models specifically calibrated for middle-aged and elderly Chinese populations constitutes a significant impediment to implementing precision medicine initiatives.

Recent years have witnessed promising developments in machine learning methodologies for cardiovascular disease risk prediction, with pertinent investigations demonstrating superior prognostic performance compared to conventional statistical models, consistently achieving area under the curve values exceeding 0.75 [5, 6]. The CHARLS represents an exceptional data resource for addressing this scientific challenge, offering high-quality longitudinal cohort information [7]. Initiated in 2011, this surveillance program encompasses 28 provinces nationwide, establishing a nationally representative repository of middle-aged and elderly individuals aged ≥ 45 years [8]. Compared with cross-sectional investigations, longitudinal cohort data facilitate the capture of temporal trajectories of risk factors, thereby enhancing long-term predictive capacity [5, 9]. Substantial progress has been achieved in cardiovascular disease prediction through machine learning applications, with ensemble learning methodologies and deep neural network architectures attaining predictive accuracies exceeding 90% [10, 11]. Nevertheless, prevailing models predominantly manifest "black-box" characteristics, lacking clinically acceptable interpretability, which significantly constrains their translational implementation [6, 12, 13]. Explainable artificial intelligence technologies predicated on methodologies such as SHAP enable quantification of individual feature contributions to predictive outcomes, providing a transparent explanatory framework for clinical decision-making [14, 15].

Given these considerations, long-term cardiovascular disease risk prediction for the Chinese middle-aged and elderly population continues to confront substantial challenges. Existing prediction models predominantly derive from Western population cohorts, with their applicability to Chinese populations remaining inadequately validated. Traditional statistical methodologies exhibit inherent limitations when processing large-scale, multidimensional longitudinal data, while machine learning technologies, despite demonstrating superior predictive performance, universally lack clinically acceptable interpretability. Accordingly, this investigation leverages comprehensive data from four waves of CHARLS surveillance spanning 2011–2020, employing multiple state-of-the-art machine learning algorithms to develop a cardiovascular disease risk prediction model specifically tailored for the Chinese middle-aged and elderly population, while implementing explainable artificial intelligence technologies to achieve transparent elucidation of model decision-making processes. This research prioritizes three fundamental elements: long-term predictive capacity, clinical utility, and interpretability, striving to enhance predictive accuracy while ensuring clinical credibility. The anticipated outcomes will not only provide scientific instrumentation for individualized cardiovascular disease prevention in the Chinese middle-aged and elderly population but also catalyze the clinical translation of artificial intelligence-driven precision medicine paradigms in cardiovascular disease prevention and management domains.

Methods

Study design and data source

This investigation employed a prospective cohort study design utilizing the CHARLS database [7]. CHARLS represents a nationally representative longitudinal surveillance program conducted by the National School of Development at Peking University, targeting middle-aged and elderly individuals aged ≥ 45 years and their spouses. The survey implements a multistage stratified probability sampling methodology, encompassing 150 counties and districts across 28 provinces in China, designed to evaluate socioeconomic and health status within community-dwelling populations. The baseline survey was completed during 2011–2012, with subsequent follow-up assessments conducted at 2–3 year intervals. This study incorporated baseline survey data (2011) and four consecutive waves of follow-up assessments (2013, 2015, 2018, and 2020), spanning a 9-year observational period. The research protocol received approval from the Peking University Institutional Review Board (IRB00001052-11015), with written informed consent obtained from all participants. Study Population and Inclusion/Exclusion Criteria.

The baseline survey enrolled 17,708 participants, from which the study population was determined through rigorous screening criteria. Exclusion criteria comprised: age < 45 years, prevalent heart disease or stroke at baseline, missing cardiovascular disease information at baseline, absence of blood biochemistry test data, and individuals with > 20% missing values for key variables. To ensure longitudinal analysis integrity, participants lost to follow-up or with missing outcome information during the surveillance period were further excluded. Given the incomplete mortality exit records in waves 3 and 4, sample sizes were adjusted accordingly at each time point. Ultimately, 9,090 participants were included in the cross-sectional analysis, with 8,080 participants completing the 9-year cardiovascular disease risk prediction analysis (Fig. 1: Flow diagram Fig. 1.1: Flow diagram).

Outcome variable definition

The primary outcome was incident cardiovascular disease (CVD), defined as cardiac events (including coronary heart disease, angina pectoris, heart failure, and other cardiac disorders) or cerebrovascular disease (stroke). Outcome ascertainment was conducted through standardised questionnaires based on participants' self-reported responses to physician-diagnosed conditions, specifically: "Has a doctor ever told you that you have heart problems, angina, coronary heart disease, heart failure, or other heart-related conditions?" and analogous enquiries regarding stroke. This definition aligns with previous cardiovascular disease investigations utilising CHARLS data and has demonstrated robust reliability across multiple validation studies [16, 17].

Rationale for composite endpoint

Despite disparate pathophysiological mechanisms underlying cardiac and cerebrovascular events, both entities share cardinal cardiovascular risk determinants (hypertension, diabetes mellitus, dyslipidaemia, tobacco consumption) and frequently co-occur or develop sequentially in middle-aged and older populations. The composite CVD endpoint confers three substantive advantages: (1) identification of individuals with elevated overall cardiovascular risk, facilitating implementation of comprehensive preventive interventions; (2) enhancement of statistical power in large-scale epidemiological investigations, ensuring adequate event accrual for model derivation and validation; (3) reflection of real-world cardiovascular disease burden, consonant with clinical imperatives for holistic cardiovascular health assessment. This definition accords with established frameworks employed in internationally recognised cohort studies, including the Framingham Heart Study, the Atherosclerosis Risk in Communities (ARIC) study, and the

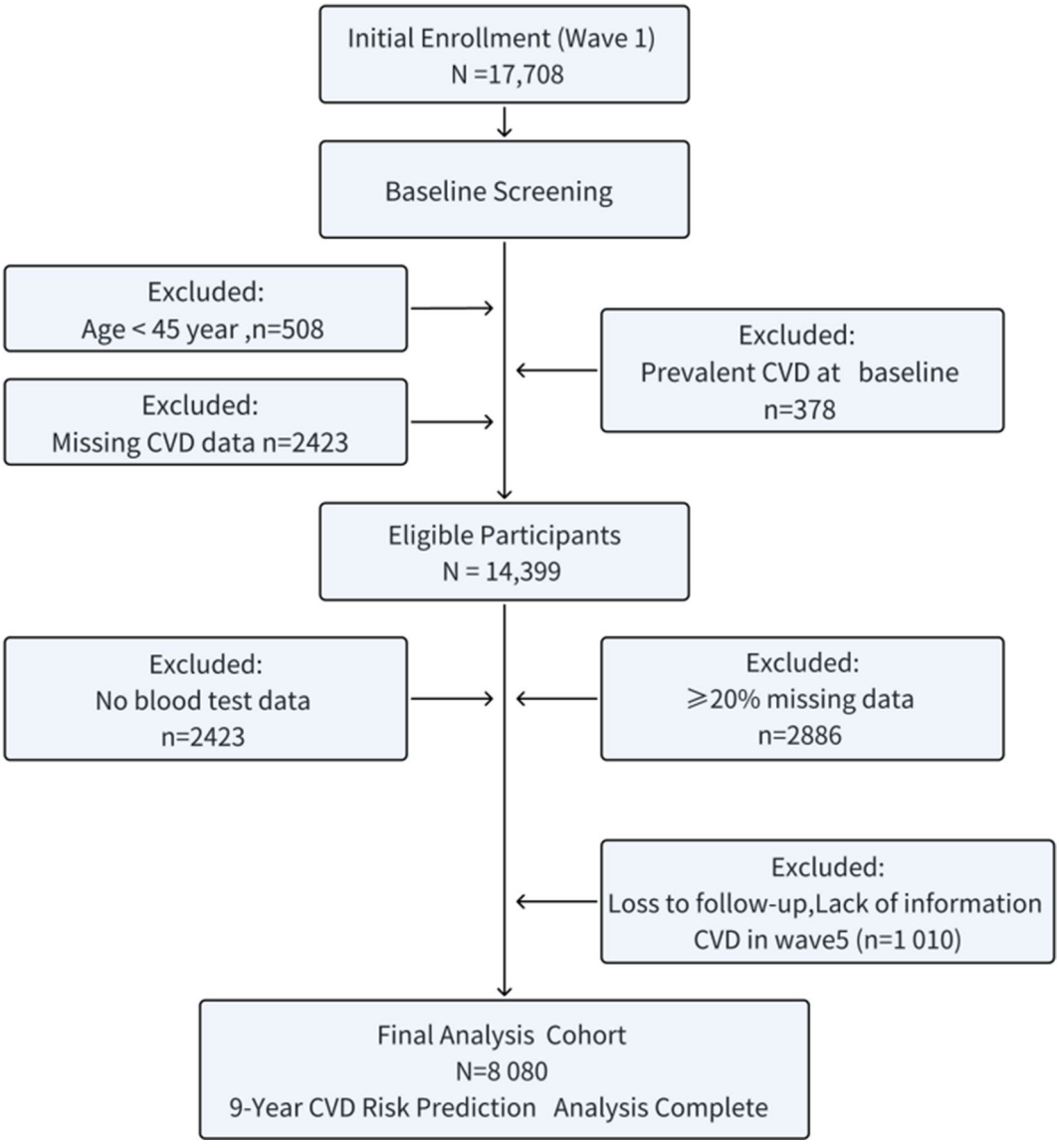


Fig. 1 Flow diagram

European Systematic Coronary Risk Evaluation (SCORE) system.

Predictor variable construction

Informed by antecedent literature and the structural characteristics of the CHARLS database, we assembled a comprehensive predictor variable set encompassing demographic attributes, functional health assessments, and laboratory biomarkers. Demographic variables comprised age, sex, ethnicity, educational attainment, residential locality, and marital status. Health status evaluation incorporated self-rated health, lifestyle behaviours (tobacco use, alcohol consumption, physical activity), activities of daily living, cognitive function scores, depressive symptomatology scales, and anthropometric measurements (height, weight, waist circumference, systolic blood pressure, diastolic blood pressure). Laboratory parameters included inflammatory markers (C-reactive protein), glycaemic indices (fasting plasma glucose, glycated haemoglobin), lipid profile components (total cholesterol, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, triglycerides), and renal function markers (creatinine, blood urea nitrogen, uric acid, cystatin C).

Additionally, several composite health metrics were computed to augment predictive capacity. Body mass index (BMI) was calculated as weight (kg) divided by height squared (m^2). Skeletal muscle mass was estimated utilising validated prediction equations calibrated for Chinese populations. Low handgrip strength was defined as < 28 kg for men or < 18 kg for women. Metabolic syndrome was diagnosed according to International

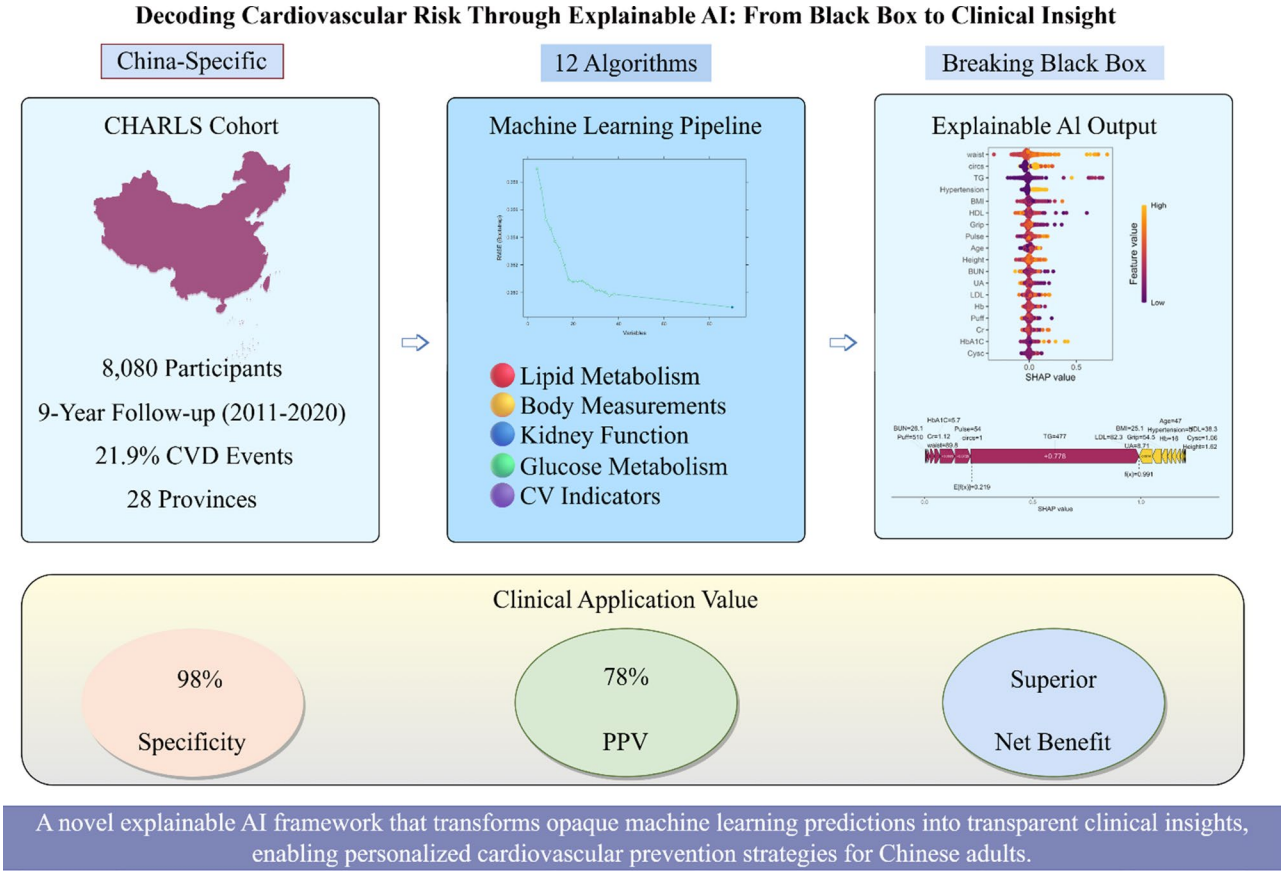


Fig. 1.1 Flow diagram. By Figdraw

Diabetes Federation criteria, requiring central adiposity (waist circumference ≥ 90 cm in men, ≥ 80 cm in women) plus at least two of the following: elevated blood pressure (≥ 130/85 mmHg or antihypertensive medication use), dysglycaemia (fasting plasma glucose ≥ 5.6 mmol/L or diabetes history), hypertriglyceridaemia (≥ 1.7 mmol/L), or reduced high-density lipoprotein cholesterol (< 1.03 mmol/L in men, < 1.29 mmol/L in women). Circadian syndrome was defined as the presence of at least four components among seven: increased waist circumference, dysglycaemia, elevated blood pressure, dyslipidaemia, insufficient sleep duration, and depressive symptoms.

Temporal validity assurance

To ensure temporal validity of predictions and preclude data leakage, all predictor variables were strictly confined to baseline (2011) measurements. Cardiovascular disease outcomes were ascertained through four successive follow-up waves (2013, 2015, 2018, and 2020), ensuring that all predictor variable measurements temporally preceded outcome occurrence. This investigation employed a single-timepoint prediction design, excluding dynamic variable changes during follow-up.

Statistical analysis methods

Data preprocessing and quality control

Addressing inevitable missing data in clinical research, we implemented multiple imputation techniques for systematic handling. Multiple imputation not only yields unbiased parameter estimates but also effectively quantifies uncertainty attributable to missing data [18–20]. Predicated on the missing at random (MAR) assumption—whereby missingness probability depends solely on observed rather than unobserved values—data imputation was executed using the "mice" package in R software. Continuous variables were expressed as mean ± SD (normal distribution) or median with interquartile range (skewed distribution) following Shapiro-Wilk normality testing, with between-group comparisons conducted using independent-samples t-tests or Mann-Whitney U tests, respectively. Categorical variables were described as frequencies (percentages), with between-group differences assessed via Pearson's χ^2 test. All statistical tests were two-tailed with significance threshold $\alpha = 0.05$.

Feature selection and model development

Recursive feature elimination (RFE) coupled with 10-fold cross-validation was employed to identify the optimal

feature subset from 90 candidate variables, determining the optimal balance between performance and complexity through performance-complexity trajectory analysis. The recursive feature elimination procedure was executed exclusively on training set baseline data, with the validation set remaining entirely independent of feature selection. To ensure clinical plausibility and temporal validity of selected features, the following constraints were imposed: (1) prioritisation of established cardiovascular risk factors (age, sex, blood pressure, lipid profile, smoking status) to ensure clinical interpretability; (2) for biomarkers potentially reflecting subclinical disease, evaluation of their evidence as independent predictors in antecedent prospective cohort investigations; (3) assessment of feature stability through cross-validation to avoid selecting unstable features performing well only in specific data partitions.

Within a multi-algorithm comparative framework, we systematically evaluated 12 machine learning algorithms: logistic regression, support vector machine, multilayer perceptron, Gaussian process, random forest, gradient boosting machine (GBM), AdaBoost, XGBoost, naïve Bayes, k-nearest neighbours, and neural networks. The dataset was partitioned into training and validation sets using stratified random sampling (7:3 ratio), with hyperparameter optimisation conducted via grid search. All feature selection and parameter tuning were executed exclusively on the training set to prevent data leakage. To mitigate overfitting, gradient boosting machine constraints included tree depth limitation (max_depth = 3–5), minimum leaf node sample size (min_samples_leaf = 20), and early stopping strategies. All preprocessing procedures (missing value imputation, standardisation) were fitted on training set parameters before application to the validation set.

Model performance evaluation

Model performance was comprehensively assessed through multidimensional metrics. Area under the receiver operating characteristic curve (AUC) served as the primary discrimination index, supplemented by classification performance metrics including sensitivity, specificity, positive predictive value, negative predictive value, and F1 score. Training set stability was evaluated via 5-fold cross-validation (gradient boosting machine cross-validated AUC 0.798 ± 0.015), with validation set performance quantified through 1000 bootstrap resampling iterations to derive 95% confidence intervals.

To evaluate model performance across diverse clinical application scenarios, classification performance was systematically analysed across threshold values ranging from 0.1 to 0.9 (0.1 increments). Youden's index (sensitivity + specificity – 1) was utilised to identify optimal statistical equilibrium between sensitivity and specificity. Performance of the Framingham cardiovascular risk score and China-PAR score was evaluated on identical validation sets. By adjusting decision thresholds across models to achieve approximately 80% and 90% specificity, we compared sensitivity between models to assess performance advantages of our model relative to conventional scoring systems. DeLong's test was employed to compare AUC differences between models, with $P < 0.05$ denoting statistical significance.

Decision curve analysis (DCA) quantified net clinical benefit across risk threshold values, providing quantitative evidence for clinical decision-making. Calibration curves assessed concordance between predicted and observed probabilities, with Hosmer-Lemeshow test evaluating calibration ($P > 0.05$ indicating satisfactory calibration).

Robustness verification

Sensitivity analyses encompassed: (1) ten iterations with different random seeds to evaluate model stability across data partitions; (2) temporal sequence validation, partitioning datasets chronologically to simulate authentic clinical application scenarios; (3) model reconstruction after excluding variables with > 30% missingness to assess impact of missing data handling strategies.

Interpretability analysis

Systematic interpretability analysis was conducted using SHAP (SHapley Additive exPlanations) methodology [21, 22]. SHAP, grounded in game-theoretic Shapley values, assigns unique importance scores to each predictor through an additive feature attribution framework, ensuring that cumulative feature contributions equal the difference between model predictions and baseline expected values, thereby providing mathematically rigorous and intuitive interpretation.

Analysis was conducted across global and local dimensions: global interpretation identified pivotal clinical indicators for cardiovascular disease risk prediction through SHAP feature importance plots, with summary plots revealing non-linear relationships and interaction effects between features and outcome variables; local interpretation generated individualised feature attribution analyses for specific patients, with waterfall plots demonstrating how each feature shifts individual predicted risk from population baseline towards personalised predictions, and dependence plots exploring feature interaction effects. This systematic approach enables clinicians to identify principal risk drivers for individual patients, facilitating transition from population-level prediction towards individualised risk management.All statistical analyses were performed using R software (version 4.2.3).

Results

Study population and baseline characteristics

The baseline survey enrolled 17,708 participants, of whom 8,080 fulfilled eligibility criteria and completed the 9-year cardiovascular disease risk prediction analysis following rigorous screening. Exclusion criteria comprised: age < 45 years, prevalent cardiac or cerebrovascular disease at baseline, > 20% missingness for key variables, and loss to follow-up during the observation period. The dataset was randomly partitioned (7:3 ratio), yielding a training cohort of 5,699 participants, among whom 1,248 (21.9%) experienced incident cardiovascular disease events.

Median age in the training cohort was 58 years (IQR 52–65), with the cardiovascular disease event group demonstrating significantly advanced age compared with the event-free group (median 59 years vs. 57 years, $p < 0.001$). Women constituted 53.3% of participants, with higher representation in the cardiovascular disease event group (56.3% vs. 52.5%, $p = 0.021$). Rural residents comprised 67.2% of the cohort, with lower rural representation observed in the cardiovascular disease event group (62.5% vs. 68.6%, $p < 0.001$).

The cardiovascular disease event group exhibited markedly elevated prevalence of cardiometabolic disorders: hypertension (40.8% vs. 16.8%), dyslipidaemia (14.7% vs. 5.2%), and diabetes mellitus (8.7% vs. 3.5%; all $p < 0.001$). Biochemical parameters revealed pronounced metabolic derangements: elevated fasting plasma glucose (5.92 vs. 5.64 mmol/L [106.56 vs. 101.52 mg/dL]), elevated triglycerides (1.52 vs. 1.12 mmol/L [134.52 vs. 99.12 mg/dL]), reduced HDL cholesterol (1.14 vs. 1.33 mmol/L [44.07 vs. 51.42 mg/dL]), and elevated C-reactive protein (1.34 vs. 0.91 mg/L; all $p < 0.001$). Anthropometric measurements demonstrated that the cardiovascular disease event group had significantly higher BMI (24.85 vs. 22.73 kg/m²), greater waist circumference (90.0 vs. 82.5 cm), and elevated systolic (132.8 vs. 124.0 mmHg) and diastolic (77.55 vs. 73.5 mmHg) blood pressure. Metabolic syndrome prevalence was substantially higher in the cardiovascular disease event group (77.9% vs. 48.8%, $p < 0.001$; Table 1).

Feature selection and predictor identification

The recursive feature elimination algorithm identified 18 optimal predictors from 90 candidate variables. The feature selection process exhibited a distinct performance saturation characteristic: as the variable count increased from four to 18, root mean square error declined from 0.3590 to 0.3509 (2.3% reduction), whilst the coefficient of determination increased from 0.2540 to 0.2859 (12.6% improvement). Beyond 18 variables, the marginal contribution of additional features diminished markedly, with the 18-variable configuration achieving 98.9% of the maximum observed coefficient of determination.

The final predictor set encompassed multiple physiological systems: lipid metabolism indices (triglycerides, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol), anthropometric parameters (waist circumference, BMI, height, handgrip strength, vital capacity), renal function biomarkers (uric acid, blood urea nitrogen, cystatin C, creatinine), glycaemic indices (glycated haemoglobin), haematological parameters (haemoglobin), cardiovascular indicators (pulse rate,

Table 1 RFE cross-validation performance across feature subsets

Variables	RMSE	Rsquared	MAE	RMSESD	RsquaredSD	MAESD
4	0.3590	0.2540	0.2572	0.007173	0.02265	0.003558
6	0.3575	0.2599	0.2502	0.006793	0.02081	0.006125
8	0.3553	0.2675	0.2545	0.007067	0.02308	0.006493
10	0.3546	0.2704	0.2537	0.006879	0.02216	0.006049
12	0.3537	0.2743	0.2527	0.006087	0.01906	0.004668
14	0.3532	0.2766	0.2547	0.006038	0.01927	0.004396
16	0.352	0.2813	0.2534	0.006195	0.01954	0.004224
18	0.3509	0.2859	0.2525	0.00549	0.01765	0.003106
20	0.3508	0.2869	0.2539	0.005215	0.01565	0.002902
22	0.3509	0.2865	0.2539	0.005437	0.01678	0.002849
24	0.3508	0.2866	0.2538	0.005486	0.01732	0.003229
26	0.3507	0.2876	0.2549	0.005268	0.01677	0.003039
28	0.3504	0.2887	0.2546	0.00521	0.01705	0.002858
30	0.3501	0.2898	0.2544	0.005366	0.01618	0.002944
32	0.3502	0.2900	0.2552	0.005506	0.01732	0.003056
34	0.3500	0.2905	0.2548	0.005314	0.01647	0.002894
36	0.3498	0.2915	0.2545	0.005417	0.01615	0.002912
38	0.3499	0.2912	0.2550	0.005335	0.0160	0.002852
90	0.3489	0.2952	0.2539	0.005554	0.01702	0.003014

hypertensive status), and demographic characteristics (age, circulatory disease history). This multidimensional feature constellation reflects the intricate pathophysiological mechanisms underlying cardiovascular disease risk (Table 1; Fig. 2).

Model performance comparison and validation

In the systematic comparison of 12 machine learning algorithms, the random forest algorithm demonstrated flawless classification performance on the training set (AUC = 1.00, accuracy = 100%), yet exhibited substantial performance deterioration on the validation set (AUC = 0.799, accuracy = 83.45%), indicating severe overfitting.

Ensemble learning methods achieved superior predictive performance whilst maintaining robust generalisability: AdaBoost attained the highest training set AUC of 0.932 (95% CI 0.926–0.939), with validation set AUC of 0.784 (95% CI 0.763–0.805); gradient boosting machine (GBM) demonstrated the most balanced performance on the validation set, with training set AUC of 0.885 and more robust validation set performance (AUC = 0.798, sensitivity = 30%, specificity = 98%, positive predictive value = 78%).

Conventional methods exhibited relatively modest performance: logistic regression achieved validation set AUC of 0.758, support vector machine 0.724, and neural network 0.750. Whilst naïve Bayes attained the highest sensitivity (training set 53%, validation set 48%), its specificity was markedly deficient (training set 87%, validation set 85%). Based on comprehensive consideration of prediction stability, generalisability, and clinical utility, GBM was designated the optimal model (Tables 2 and 3; Fig. 3).

GBM demonstrated stable performance in 5-fold cross-validation with AUC of 0.798 ± 0.015 (range 0.781–0.816). Random forest yielded training set AUC = 1.00 but cross-validation AUC of only 0.756 ± 0.022, confirming overfitting. Bootstrap analysis revealed validation set AUC of 0.805 (95% CI 0.785–0.825). Across ten random partitions, the AUC coefficient of variation was 2.1% (range 0.792–0.818). Temporal sequence validation yielded AUC of 0.789. After excluding high-missingness variables, AUC remained at 0.801 ($p = 0.43$).

Threshold optimisation and comparison with existing models

Systematic threshold analysis identified optimal decision points for distinct clinical scenarios: at threshold 0.2, Youden's index reached its maximum value of 0.431, corresponding to sensitivity 70.9% and specificity 72.2%. At threshold 0.1, sensitivity attained 98.3%, though specificity declined to 26% with positive predictive value of

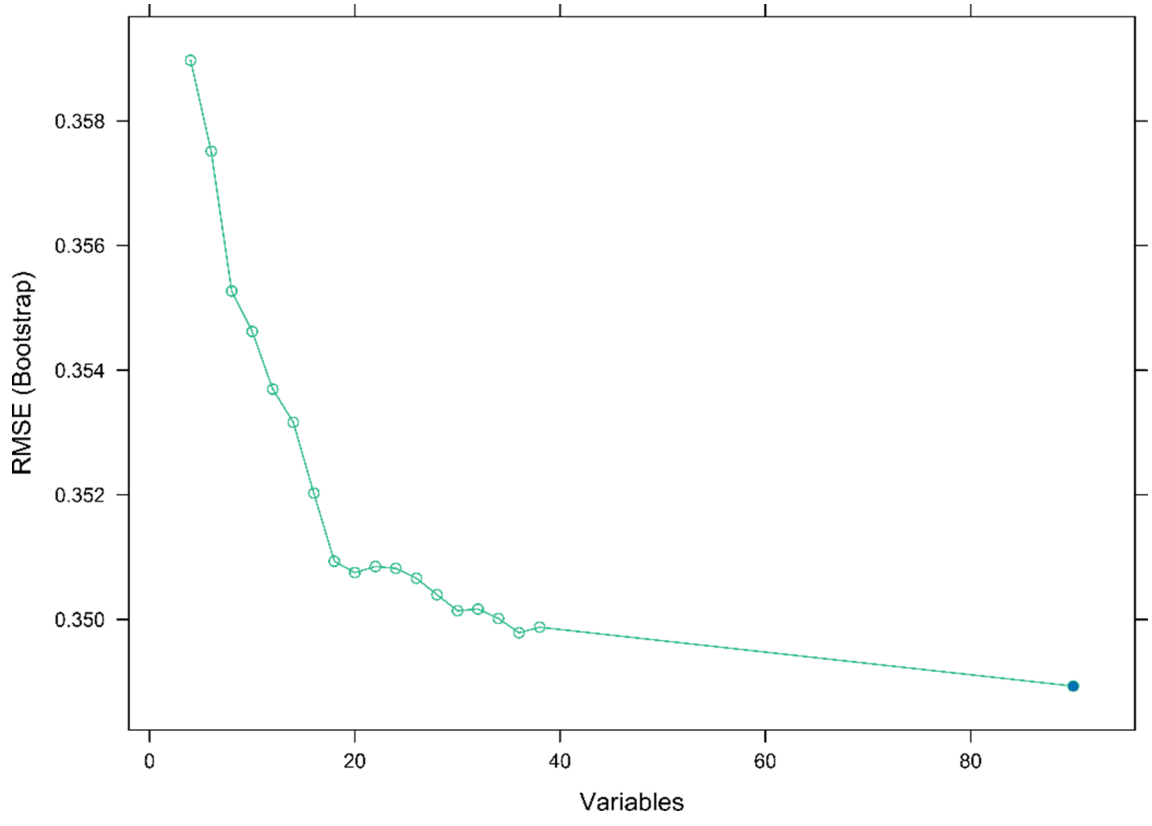


Fig. 2 Performance metrics of recursive feature elimination across different feature subset sizes

Table 2 Performance comparison of machine learning algorithms on training dataset

CLS	Sensitivity	Specificity	Pos Pred Value	Neg Pred Value	Recall	F1	Accuracy	Auc
Logistic Regression	0.30 (0.28, 0.33)	0.95 (0.95, 0.96)	0.65 (0.61, 0.68)	0.83 (0.82, 0.84)	0.3045	0.4139	0.8112(0.8008, 0.8213)	0.791(0.7777–0.8053)
Neural Network	0.23 (0.20, 0.25)	0.98 (0.98, 0.99)	0.80 (0.76, 0.84)	0.82 (0.81, 0.83)	0.2268	0.3538	0.8186(0.8083, 0.8285)	0.802(0.7883–0.8157)
svm	0.22 (0.18, 0.25)	0.97 (0.96, 0.98)	0.67 (0.59, 0.74)	0.81 (0.79, 0.83)	0.2166	0.3272	0.8013(0.7847, 0.8172)	0.762(0.747–0.7778)
Multilayer Perceptron	0.27 (0.24, 0.29)	0.98 (0.97, 0.98)	0.75 (0.71, 0.79)	0.83 (0.82, 0.84)	0.2652	0.3919	0.8198(0.8096, 0.8297)	0.798(0.7836–0.8118)
Gaussian Process	0.32 (0.30, 0.35)	0.97 (0.96, 0.97)	0.75 (0.71, 0.78)	0.84 (0.83, 0.85)	0.3245	0.4528	0.8282(0.8182, 0.8379)	0.822(0.8087–0.8352)
GBM	0.40 (0.38, 0.43)	0.99 (0.98, 0.99)	0.89 (0.86, 0.91)	0.86 (0.85, 0.86)	0.4046	0.5562	0.8586(0.8493, 0.8675)	0.885(0.8755–0.8954)
Naive Bayes	0.53 (0.50, 0.56)	0.87 (0.85, 0.88)	0.52 (0.50, 0.55)	0.87 (0.86, 0.88)	0.5288	0.5263	0.7915(0.7808, 0.802)	0.811(0.7984–0.8243)
adaboost	0.58 (0.55, 0.61)	0.97 (0.96, 0.97)	0.84 (0.81, 0.86)	0.89 (0.88, 0.90)	0.5809	0.6856	0.8833(0.8747, 0.8915)	0.932(0.9256–0.9391)
XGB	0.33 (0.30, 0.36)	0.99 (0.98, 0.99)	0.87 (0.84, 0.90)	0.84 (0.83, 0.85)	0.3285	0.4776	0.8426(0.8329, 0.852)	0.860(0.8485–0.8706)
Random Forest	1.00 (1.00, 1.00)	1.00 (1.00, 1.00)	1.00 (1.00, 1.00)	1.00 (1.00, 1.00)	1.00	1.00	1.00 (1.00, 1.00)	1.00 (1.00, 1.00)
Decision Tree	0.44 (0.41, 0.47)	0.95 (0.94, 0.95)	0.69 (0.66, 0.72)	0.86 (0.85, 0.87)	0.4391	0.5373	0.8344(0.8244, 0.8439)	0.843(0.8313–0.8544)
KNN	0.21 (0.19, 0.23)	0.98 (0.98, 0.99)	0.79 (0.74, 0.84)	0.82 (0.81, 0.83)	0.2083	0.3299	0.8147(0.8044, 0.8247)	0.823(0.7984–0.8243)

Table 3 Performance comparison of machine learning algorithms on validation dataset

CLS	Sensitivity	Specificity	Pos Pred Value	Neg Pred Value	Recall	F1	Accuracy	Auc
Logistic Regression	0.29 (0.25, 0.33)	0.95 (0.94, 0.96)	0.64 (0.57, 0.70)	0.82 (0.81, 0.84)	0.2881	0.3969	0.8047(0.7882, 0.8205)	0.758(0.7351–0.7805)
Neural Network	0.17 (0.14, 0.20)	0.98 (0.98, 0.99)	0.73 (0.64, 0.80)	0.80 (0.79, 0.82)	0.1657	0.2699	0.8001(0.7834, 0.816)	0.750(0.7264–0.7726)
svm	0.22 (0.18, 0.25)	0.97 (0.96, 0.98)	0.67 (0.59, 0.74)	0.81 (0.79, 0.83)	0.2166	0.3272	0.8013(0.7847, 0.8172)	0.724(0.699–0.7488)
Multilayer Perceptron	0.20 (0.16, 0.23)	0.97 (0.96, 0.98)	0.64 (0.56, 0.72)	0.80 (0.78, 0.82)	0.1959	0.3001	0.7963(0.7796, 0.8123)	0.751(0.7278–0.7741)
Gaussian Process	0.27 (0.23, 0.31)	0.96 (0.95, 0.97)	0.64 (0.57, 0.70)	0.82 (0.80, 0.84)	0.2655	0.3755	0.8030(0.7865, 0.8188)	0.761(0.7382–0.7842)
GBM	0.30 (0.26, 0.34)	0.98 (0.97, 0.98)	0.78 (0.72, 0.83)	0.83 (0.81, 0.84)	0.2994	0.4327	0.8249(0.809, 0.8399)	0.798(0.7777–0.8191)
Naive Bayes	0.48 (0.44, 0.53)	0.85 (0.83, 0.87)	0.48 (0.44, 0.52)	0.85 (0.83, 0.87)	0.4840	0.4826	0.7686(0.7511, 0.7854)	0.760(0.7368–0.7825)
adaboost	0.37 (0.33, 0.41)	0.93 (0.92, 0.94)	0.61 (0.55, 0.66)	0.84 (0.82, 0.85)	0.3672	0.4577	0.8060(0.7895, 0.8217)	0.784(0.7626–0.8047)
XGB	0.27 (0.23, 0.31)	0.99 (0.98, 0.99)	0.85 (0.78, 0.90)	0.82 (0.81, 0.84)	0.2693	0.4086	0.8261(0.8103, 0.8411)	0.801(0.7805–0.8215)
Random Forest	0.30 (0.26, 0.34)	0.99 (0.98, 0.99)	0.87 (0.82, 0.92)	0.83 (0.81, 0.85)	0.3013	0.4482	0.8345(0.819, 0.8492)	0.799(0.7772–0.821)
Decision Tree	0.39 (0.35, 0.44)	0.94 (0.93, 0.95)	0.65 (0.59, 0.70)	0.84 (0.83, 0.86)	0.3936	0.4889	0.8165(0.8003, 0.8318)	0.788(0.7665–0.8092)
KNN	0.17 (0.14, 0.20)	0.98 (0.97, 0.98)	0.66 (0.57, 0.74)	0.80 (0.79, 0.82)	0.1676	0.2673	0.795(0.7783, 0.8111)	0.743(0.7195–0.7668)

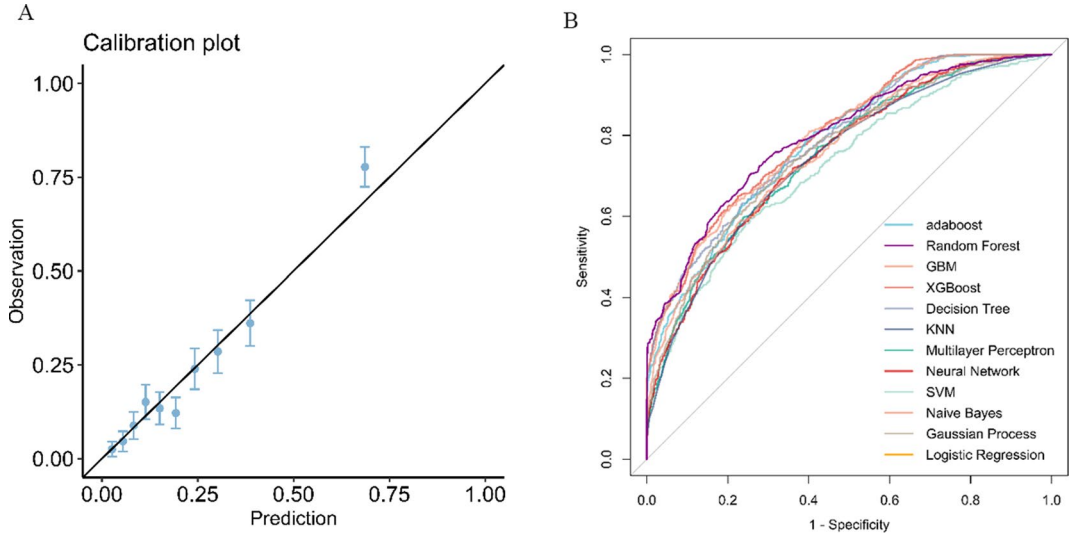


Fig. 3 Receiver operating characteristic curves for twelve machine learning algorithms across training and validation datasets

Table 4 Performance metrics of the gradient boosting machine model at different prediction thresholds

threshold	sensitivity	specificity	ppv	npv	f1 score	Youden index
0.1	98.3	26.0	27.2	98.2	0.427	0.243
0.2	70.9	72.2	41.8	89.8	0.526	0.431
0.3	47.7	87.9	52.7	85.6	0.500	0.356
0.4	36.0	94.6	65.1	84.0	0.464	0.306
0.5	25.5	98.7	84.5	82.4	0.392	0.242
0.6	21.8	99.8	96.7	81.9	0.355	0.216
0.7	20.6	99.8	97.3	81.7	0.341	0.205
0.8	18.4	99.8	97.0	81.3	0.309	0.182
0.9	4.9	99.9	96.3	78.8	0.093	0.048

PPV, positive predictive value; NPV, negative predictive value; F1 score, harmonic mean of precision and recall; Youden index = sensitivity + specificity − 1

merely 27.2%. At the default threshold of 0.5, specificity reached 98.7% with positive predictive value 84.5%, yet sensitivity was only 25.5% (Table 4).

Comparison with existing risk scores demonstrated that the GBM model achieved AUC of 0.798 (95% CI 0.785–0.825), significantly exceeding the Framingham Risk Score (0.638, 95% CI 0.611–0.665) and China-PAR score (0.654, 95% CI 0.627–0.680). At matched specificity of approximately 80%, the present model's sensitivity was 61.7%, substantially superior to the Framingham score (38.6%) and China-PAR score (40.0%). At approximately 90% specificity, the present model's sensitivity was 44.3%, likewise exceeding the Framingham score (23.8%) and China-PAR score (30.4%).

Calibration performance and clinical decision value

The GBM model exhibited satisfactory predictive calibration characteristics. Calibration curve analysis revealed substantial concordance between predicted probabilities and observed frequencies, with optimal performance in the intermediate risk range (0.2–0.75). Modest deviations were observed at probability extremes: when predicted probability approached 1.0, the model demonstrated slight overconfidence, whilst marginal underestimation was present in the low-probability range approaching zero (Fig. 4A).

Decision curve analysis confirmed the broad clinical applicability of the GBM model. The model-derived risk stratification strategy generated positive net benefit across threshold probabilities ranging from 0 to 0.95, systematically surpassing the no-intervention baseline. The treat-all strategy produced positive net benefit only at extremely low thresholds (< 0.25), subsequently declining precipitously into negative territory. Beyond threshold 0.05, the model strategy's net benefit consistently exceeded treat-all, with the differential widening progressively and achieving maximum separation at 0.25 (Fig. 4B).

Interpretability analysis

SHAP analysis elucidated the core drivers of the prediction model. Waist circumference, triglycerides, and hypertension history ranked among the top three features by importance, with mean absolute SHAP values

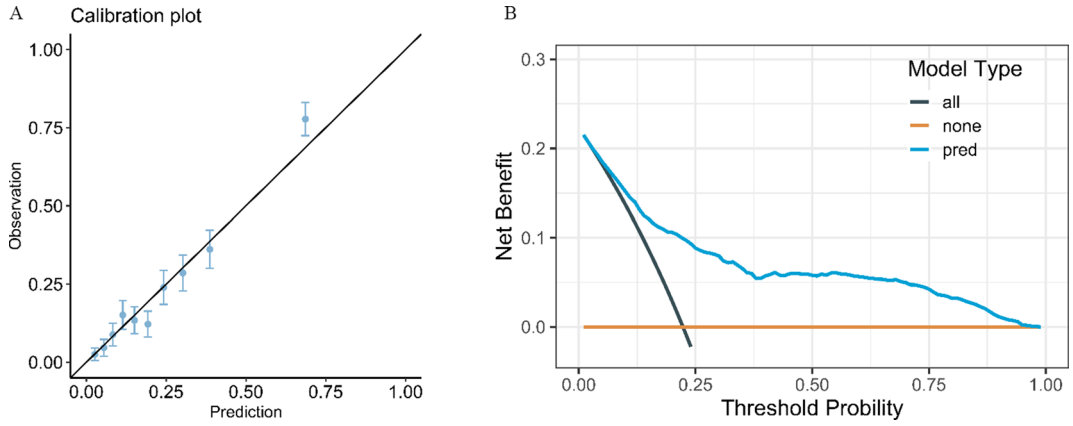


Fig. 4 Comprehensive evaluation of the gradient boosting machine (GBM) model performance. (**A**) Calibration plot demonstrating the agreement between predicted probabilities and observed frequencies across the full risk spectrum. The diagonal line represents perfect calibration, with data points indicating actual model performance. (**B**) Decision curve analysis illustrating net benefit across threshold probabilities (0–1.0). The "pred" line represents the GBM model-guided strategy, "all" indicates universal intervention, and "none" denotes no intervention baseline. The model demonstrates superior net benefit across nearly the entire range of clinically relevant decision thresholds

substantially exceeding other variables. The summary plot revealed distinct dose-response relationships: samples with elevated waist circumference concentrated in the positive SHAP value region, whilst low waist circumference samples clustered in the negative domain, forming a gradient distribution pattern from lower-left to upper-right (Fig. 5).

Individual risk attribution analysis demonstrated risk decomposition patterns across four representative samples:

Intermediate-risk sample (f(x) = 0.122): this individual's predicted risk fell below the baseline value (E[f(x)] = 0.219). Although blood urea nitrogen (15.6 mg/dL) and LDL cholesterol (131 mg/dL) conferred modest positive contributions (+ 0.0164, + 0.0266), protective factors including BMI (19.4 kg/m^2) and age (56 years) generated stronger negative contributions, ultimately reducing risk to 0.122.

Extremely high-risk sample (f(x) = 0.991): standardised waist circumference (0.778) contributed a SHAP value of + 0.0778, severely elevated triglycerides (477 mg/dL) contributed + 0.0729, and elevated blood urea nitrogen (26.1 mg/dL) further amplified risk. The cumulative effects of these factors substantially elevated the prediction from baseline to near 1.0.

Waist-circumference-dominant high-risk sample (f(x) = 0.866): waist circumference of 102 cm independently contributed a SHAP value of + 0.193, constituting the predominant portion of total risk increment. Reduced handgrip strength (12.9 kg) contributed + 0.0903, whilst HbA1c (5.7%), approaching the prediabetes threshold, contributed + 0.0793.

Low-risk sample (f(x)=−0.0393): exceptionally low triglyceride concentration (45.1 mg/dL) generated a negative SHAP value of − 0.166, representing the largest singular protective contribution observed. Optimal BMI (23.8 kg/m^2) and favourable HDL cholesterol (57.2 mg/dL) synergistically provided additional protection, yielding final risk substantially below the population mean (Fig. 6).

Across all sample analyses, the baseline value remained constant (E[f(x)] = 0.219), with the algebraic sum of individual feature SHAP values precisely equalling the final predicted value, validating the mathematical rigour of the additive decomposition framework.

Discussion

This study leveraged 9-year follow-up data from the nationally representative CHARLS cohort to successfully develop a machine learning prediction model for cardiovascular disease in middle-aged and older adults, achieving a validation set AUC of 0.798, which significantly exceeded conventional logistic regression (AUC = 0.758). This performance is comparable to recent international studies using similar methodologies, such as the LightGBM model (AUC = 0.818) based on the same dataset [23], confirming the superiority of machine learning approaches for cardiovascular risk prediction in Chinese populations.

The principal innovation of this study lies in the organic integration of high-performance prediction with clinical interpretability. Whilst traditional machine learning models demonstrate robust predictive accuracy, their "black box" characteristics severely constrain clinical implementation [24]. The SHAP framework employed herein is grounded in game-theoretic Shapley values, providing mathematically rigorous quantification of each feature's contribution and enabling multilevel interpretability from global to individual perspectives. This methodological innovation holds substantial implications for

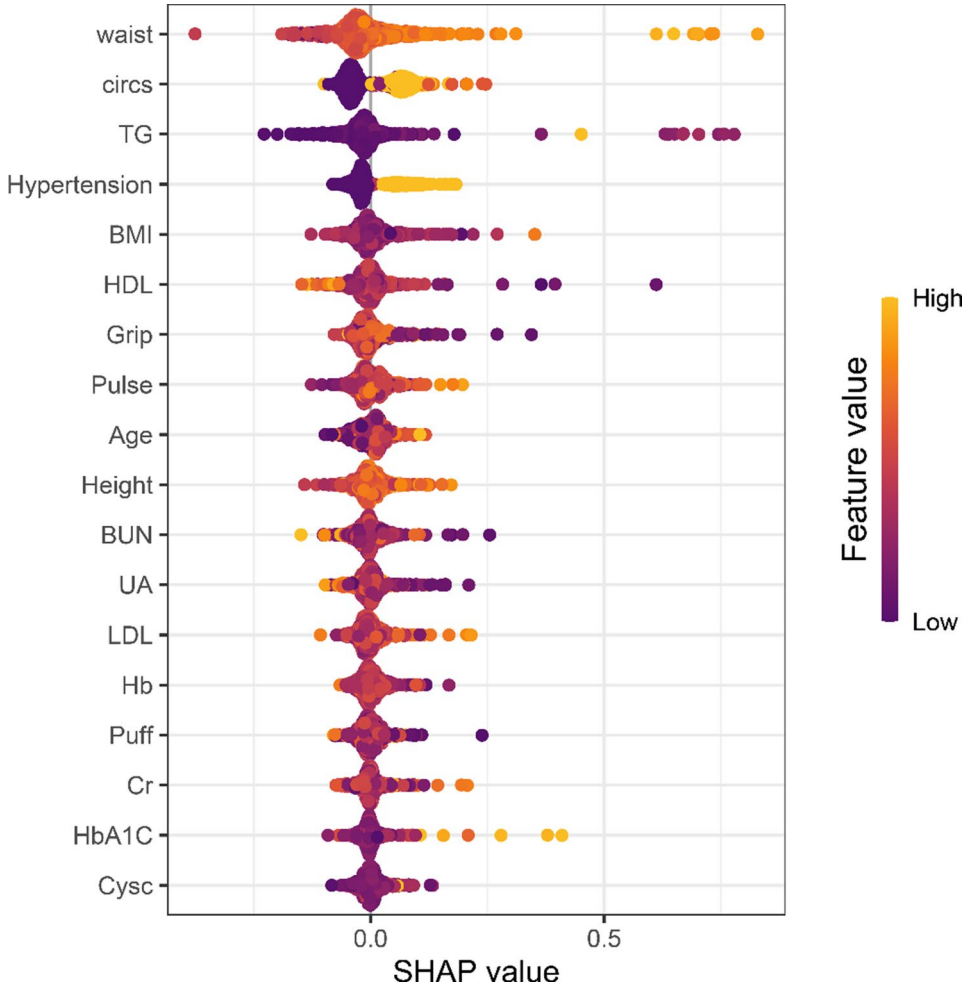


Fig. 5 Global feature importance analysis of the gradient boosting machine model. Variables are ordered by their relative contribution to predictive performance, with higher scores indicating greater influence on model output. Circs: Circadian syndrome; TG: Triglycerides; BMI: Body mass index; HDL: High Density Lipoprotein; BUN: Blood urea nitrogen; UA: Uric acid; LDL: Low Density Lipoprotein; Hb: Hemoglobin; Cr: Creatinine; HbA1c: Glycated hemoglobin; Cysc: Cystatin C

advancing translational applications of artificial intelligence in cardiovascular medicine.

SHAP analysis identified waist circumference as the primary predictor, a finding with compelling biological plausibility. As a direct measure of central adiposity, waist circumference reflects visceral adipose tissue distribution, which is intimately associated with insulin resistance, chronic inflammation, and atherosclerosis [25, 26]. A 2022 prospective cohort study demonstrated that waist circumference predicts cardiovascular mortality more effectively than BMI, corroborating the importance of waist circumference in our model [26]. Nevertheless, it is imperative to emphasise that the substantial contribution of waist circumference in the prediction model reflects its statistical predictive value rather than direct causality.

The identification of triglycerides as the second most important predictor aligns closely with current understanding of lipid metabolism's role in cardiovascular disease. Triglyceride-rich lipoproteins and their remnant particles not only participate in the pathophysiology of atherosclerosis but are also implicated in thrombotic events [27]. Partial dependence plots revealed a sharp escalation in triglycerides' impact on predicted outcomes within the 150–200 mg/dL range, a threshold below the traditional guideline-defined abnormal value (≥ 200 mg/dL). This non-linear association pattern suggests that cardiovascular risk associated with triglycerides might manifest at lower concentrations; however, this observation requires validation through dedicated causal inference studies to establish clinical intervention significance.

The importance of renal function indices in the model reflects the pivotal role of the cardiorenal axis in cardiovascular disease pathogenesis. The threshold effect at eGFR 99.3 mL/min/1.73 m^2 accords with the 2024 KDIGO guidelines emphasising combined staging of eGFR and albuminuria [28, 29]. The aggregate predictive

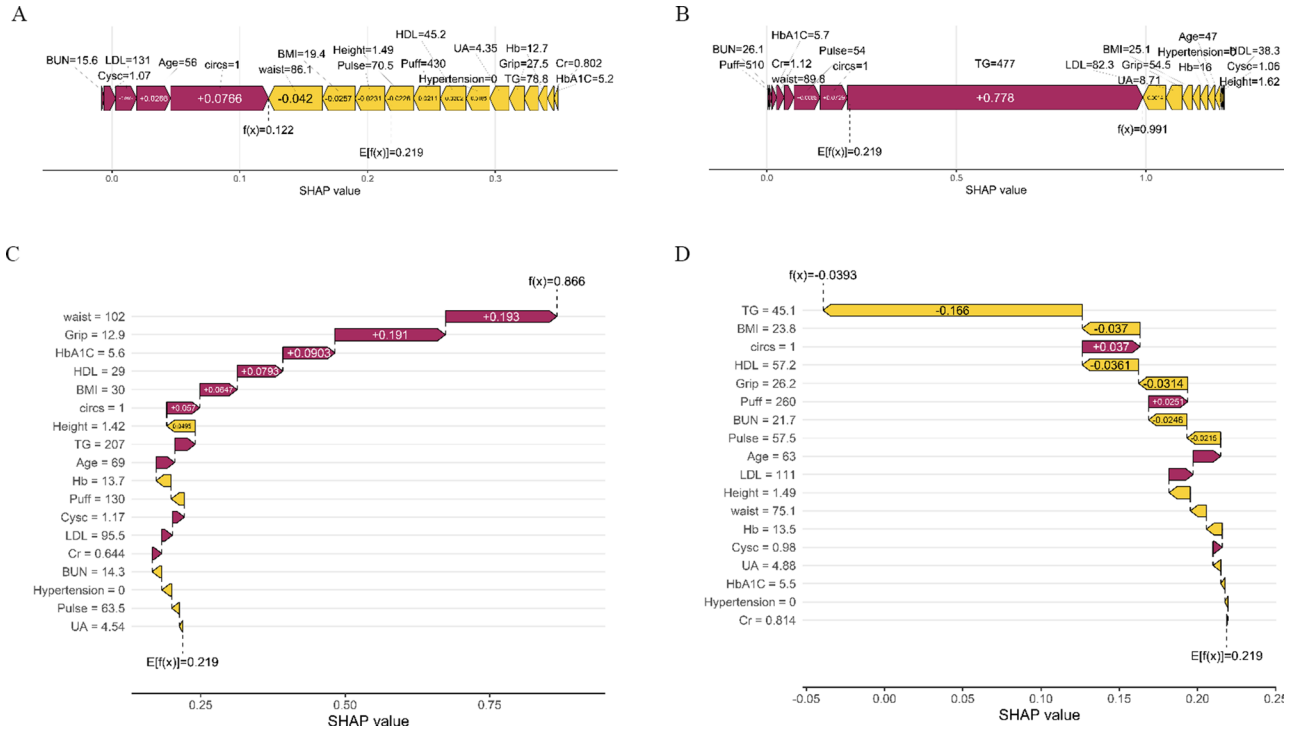


Fig. 6 Waterfall plot of individual feature attributions showing the cumulative effect of each variable on the patient's predicted risk. Features are ordered by absolute impact magnitude, enabling identification of primary risk drivers for personalized intervention strategies

contribution of serum creatinine, blood urea nitrogen, and eGFR (totalling 26.6%) underscores the value of renal function assessment in cardiovascular risk stratification. As the terminal metabolite of purine metabolism, uric acid's predictive value might reflect complex interactions amongst metabolic syndrome components, though its causal role remains contentious.

Individual SHAP analysis revealed substantial heterogeneity in risk factor contributions, which holds important implications for precision medicine. Among high-risk individuals, waist circumference alone can contribute a SHAP value as high as +0.0778, whilst in low-risk individuals, optimal triglyceride concentrations confer a protective effect of −0.0661. This individualised risk decomposition transcends the population-averaged effects of traditional risk scores, providing more granular information for clinical decision-making.

The moderating effects of sex and age on risk factor contributions further exemplify the value of individualised assessment. The waist circumference–triglyceride interaction was more pronounced in men (+ 0.042), whereas in women the interaction between hypertension history and diabetes was stronger (+ 0.038). These differences might relate to sex-specific variations in sex hormones, metabolic patterns, and disease susceptibility [30, 31]. However, these interaction effects reflect statistical associations within the prediction model and should not be directly interpreted as biological mechanisms or intervention targets.

The model's performance characteristics across different thresholds afford flexibility for clinical application. The high specificity (98.7%) at the default threshold of 0.5 renders it suitable for precise identification of high-risk populations, whilst the balanced performance at threshold 0.2 (sensitivity 70.9%, specificity 72.2%) is more appropriate for initial screening. Compared with the Framingham Risk Score (AUC = 0.638) and China-PAR score (AUC = 0.654), the present model demonstrates substantial superiority, achieving higher sensitivity at matched specificity.

Decision curve analysis confirmed that the model generates positive net benefit across the threshold probability range of 0–0.95, systematically outperforming both treat-all and no-intervention strategies. This broad clinical utility range indicates the model's applicability across diverse healthcare resource allocations and risk preferences. Nevertheless, selection of the optimal threshold requires prospective validation in real-world clinical settings, with comprehensive consideration of sensitivity–specificity trade-offs, resource constraints, and intervention cost-effectiveness.

SHAP analysis substantially enhanced the clinical acceptability of the machine learning model, which holds important implications for broader implementation of artificial intelligence in healthcare. Interpretability

not only strengthens clinician confidence in the model but also provides necessary transparency for regulatory approval and ethical review [32, 33]. Through visualisation techniques including waterfall plots and force-directed graphs, this study intuitively demonstrated individual patients' risk attribution pathways. This quantitative decomposition renders complex prediction processes comprehensible and actionable.

However, the fundamental limitations of interpretability techniques must be explicitly acknowledged. SHAP values quantify features' statistical contributions within the prediction model rather than biological causal relationships or intervention effects. This distinction is crucial: predictive importance does not equate to causal relevance, and high SHAP values do not imply that a factor is a direct disease cause or ideal intervention target. As an approximation method, interpretability might omit important information about black-box model mechanisms, necessitating cautious interpretation in clinical applications [34].

Recommending specific clinical interventions based on SHAP values risks conflating predictive importance with causal effects. For instance, whilst waist circumference contributes most substantially to the model, this does not permit direct inference that reducing waist circumference will necessarily decrease cardiovascular risk, as waist circumference might serve as a proxy variable for other underlying causal factors. Any intervention recommendations must be grounded in evidence from randomised controlled trials or rigorous causal inference studies.

The present model was developed in adults aged ≥ 45 years (median 58, IQR 52–65) using CHARLS probability sampling across 28 provinces. Generalizability to younger populations (< 45 years) requires validation, as early-onset cardiovascular events often reflect distinct pathophysiology including familial dyslipidaemia and genetic cardiomyopathies rather than cumulative age-related risk. Additional considerations include potential underrepresentation of ethnic minorities (5% vs. 8–9% nationally) and highly urbanized populations, reliance on self-reported rather than adjudicated cardiovascular endpoints, and regional heterogeneity in diagnostic capabilities affecting risk factor detection and event ascertainment. External validation in independent Chinese cohorts—including the China Kadoorie Biobank for broader geographic representation and hospital registries with adjudicated outcomes—is essential to establish performance equity across demographic subgroups and healthcare settings before widespread clinical deployment.

This study has inherent limitations of observational research. First, the static model constructed from single-timepoint baseline data fails to capture dynamic changes in risk factors. Cardiovascular risk is a dynamic process, and changes in risk factors during follow-up might substantially affect predictive accuracy. Whilst dynamic risk assessment might provide more precise predictions, it necessitates repeated evaluations, increasing implementation complexity and costs.

Second, whilst use of a composite endpoint enhanced statistical power, it constrained predictive precision for specific disease subtypes. Myocardial infarction, stroke, and heart failure possess distinct pathophysiological mechanisms and risk factor profiles; a unified prediction model might not adequately reflect this heterogeneity.

Third, self-reported cardiovascular events are susceptible to recall bias and underreporting, particularly for asymptomatic or minor events. Although such bias might underestimate true incidence, for prevention-oriented public health applications, focusing on clinically manifest events better aligns with practical needs.

Finally, the applicability of this model, developed in a Chinese population, to other ethnic and geographical populations requires further validation. Differences in genetic background, lifestyle, and environmental factors might influence the model's generalisability.

This study opens new avenues for precision cardiovascular disease prevention. First, multidimensional prediction models integrating multi-omics data (genomics, metabolomics, proteomics) promise to further enhance predictive accuracy and biological interpretability. Second, developing dynamic risk assessment models that incorporate longitudinal measurement data and time-dependent covariates might enable more precise individualised risk prediction.

At the methodological level, advances in causal machine learning provide novel tools for transitioning from predictive associations to causal inference. Combining instrumental variables, regression discontinuity, and other causal inference methods might identify genuine causal risk factors, furnishing scientific evidence for precision interventions.

Regarding clinical translation, prospective validation studies in real-world healthcare settings are needed to evaluate the model's actual clinical utility and cost-effectiveness. Simultaneously, developing user-friendly clinical decision support systems that integrate risk prediction, interpretability analysis, and evidence-based intervention recommendations will facilitate clinical implementation.

Contribution to precision cardiovascular medicine

This study represents an important transition from traditional "one-size-fits-all" risk assessment toward individualised precision evaluation. By integrating high-performance machine learning with interpretability techniques, we not only achieved accurate risk prediction but,

more importantly, provided quantitative decomposition of risk factor contributions, establishing a foundation for individualised intervention strategy formulation.

However, achieving genuine precision cardiovascular medicine requires balancing technological innovation with ethical considerations. Whilst advancing artificial intelligence clinical applications, we must ensure algorithmic fairness, transparency, and interpretability, avoiding exacerbation of health inequalities. Concurrently, establishing robust regulatory frameworks and quality control systems is essential to ensure the safety and efficacy of artificial intelligence tools.

Conclusions

This study successfully developed a high-performance and interpretable cardiovascular disease prediction model, providing important methodological innovation for precision risk assessment. SHAP analysis elucidated predictive contribution patterns of risk factors and individual heterogeneity, enhancing the model's clinical interpretability. However, a fundamental distinction exists between predictive importance and causality; any clinical application based on prediction models must proceed with full appreciation of this limitation. Future research should prioritise prospective clinical validation, dynamic risk modelling, and integration of causal inference methods to genuinely achieve translation from statistical prediction to precision intervention, ultimately improving cardiovascular disease prevention and management outcomes.

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Author contributions
ZXY and PXY conceptualized the study design and developed the machine learning framework. ZXY, LW, and YGL performed the data preprocessing, feature engineering, and statistical analyses. PXY implemented the machine learning algorithms and SHAP interpretability analysis. LW validated the model performance and conducted the comparative analysis of different algorithms. PXY drafted the initial manuscript with contributions from ZXY. ZXY provided critical revision of the methodology and supervised the overall research direction. All authors reviewed, edited, and approved the final manuscript for publication.

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Data availability
The CHARLS dataset can be downloaded from the CHARLS homepage at http://charls.pku.edu.cn/en.

Declarations

Ethical approval
The baseline and follow-up surveys of CHARLS were approved by the Peking University Ethics Review Committee (IRB00001052-11015). The study was conducted in accordance with the ethical principles of the Declaration of Helsinki. All participants signed informed consent forms.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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